

Assignment No 1

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Subject : ADS Lab

Problem Statement: Assignment based on implementation of Artificial Neural Network (ANN) using TensorFlow/Keras for prediction problems, Exemplars: Student Performance Prediction, House Price Prediction, Employee Attrition Prediction

Implementation:

```
# Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_absolute_error, mean_squared_error

import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

# -----
# Load Dataset
# -----

housing = fetch_california_housing(as_frame=True)

df = housing.frame

print(df.head())
print(df.info())

# -----
# Feature and Target
# -----

X = df.drop("MedHouseVal", axis=1)
y = df["MedHouseVal"]

# -----
# Train Test Split
```

```

# -----

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

# -----
# Feature Scaling
# -----

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# -----
# Build ANN Model
# -----

model = Sequential()

model.add(Dense(64,
                activation='relu',
                input_shape=(X_train.shape[1],)))

model.add(Dense(32,
                activation='relu'))

model.add(Dense(16,
                activation='relu'))

model.add(Dense(1,
                activation='linear'))

# -----
# Compile Model
# -----

model.compile(
    optimizer='adam',
    loss='mse',
    metrics=['mae']
)

```

```

# -----
# Train Model
# -----

history = model.fit(
    X_train,
    y_train,
    validation_split=0.2,
    epochs=50,
    batch_size=32,
    verbose=1
)

# -----
# Evaluate Model
# -----

loss, mae = model.evaluate(X_test, y_test)

print("Test Loss :", loss)
print("Test MAE :", mae)

# -----
# Prediction
# -----

y_pred = model.predict(X_test)

rmse = np.sqrt(mean_squared_error(y_test, y_pred))

print("RMSE :", rmse)

# -----
# Plot Loss
# -----

plt.plot(history.history['loss'], label='Training Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.legend()
plt.show()

```

Output:

```
MedInc HouseAge AveRooms AveBedrms Population AveOccup Latitude
Longitude MedHouseVal
```

```
0 8.3252 41.0 6.984127 1.023810 322.0 2.555556 37.88 -122.23
4.526
1 8.3014 21.0 6.238137 0.971880 2401.0 2.109842 37.86 -122.22
3.585
2 7.2574 52.0 8.288136 1.073446 496.0 2.802260 37.85 -122.24
3.521
3 5.6431 52.0 5.817352 1.073059 558.0 2.547945 37.85 -122.25
3.413
4 3.8462 52.0 6.281853 1.081081 565.0 2.181467 37.85 -122.25
3.422
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 20640 entries, 0 to 20639
```

```
Data columns (total 9 columns):
```

```
# Column Non-Null Count Dtype
```

```
--- ----
```

```
0 MedInc 20640 non-null float64
1 HouseAge 20640 non-null float64
2 AveRooms 20640 non-null float64
3 AveBedrms 20640 non-null float64
4 Population 20640 non-null float64
5 AveOccup 20640 non-null float64
6 Latitude 20640 non-null float64
7 Longitude 20640 non-null float64
8 MedHouseVal 20640 non-null float64
```

```
dtypes: float64(9)
```

```
memory usage: 1.4 MB
```

```
None
```

```
Epoch 1/50
```

```
413/413 ----- 2s 2ms/step - loss: 0.8055
```

```
- mae: 0.6101 - val_loss: 0.4308 - val_mae: 0.4804
```

```
Epoch 2/50
```

```
413/413 ----- 1s 1ms/step - loss: 0.3786
```

```
- mae: 0.4384 - val_loss: 0.3947 - val_mae: 0.4433
```

```
Epoch 3/50
```

```
413/413 ----- 1s 1ms/step - loss: 0.3556
```

```
- mae: 0.4244 - val_loss: 0.3777 - val_mae: 0.4238
```

```
Epoch 4/50
```

```
413/413 ----- 1s 1ms/step - loss: 0.3430
```

```
- mae: 0.4162 - val_loss: 0.3870 - val_mae: 0.4324
```

```
Epoch 5/50
```

```
413/413 ----- 1s 1ms/step - loss: 0.3412
```

```
- mae: 0.4110 - val_loss: 0.3844 - val_mae: 0.4167
```

```
Epoch 6/50
```

413/413 ————— 1s 1ms/step - loss: 0.3223
- mae: 0.4000 - val_loss: 0.3407 - val_mae: 0.4020
Epoch 7/50

413/413 ————— 1s 1ms/step - loss: 0.3171
- mae: 0.3946 - val_loss: 0.3500 - val_mae: 0.4011
Epoch 8/50

413/413 ————— 1s 1ms/step - loss: 0.3079
- mae: 0.3875 - val_loss: 0.3387 - val_mae: 0.4081
Epoch 9/50

413/413 ————— 1s 1ms/step - loss: 0.3019
- mae: 0.3836 - val_loss: 0.3314 - val_mae: 0.3874
Epoch 10/50

413/413 ————— 1s 1ms/step - loss: 0.2997
- mae: 0.3803 - val_loss: 0.3259 - val_mae: 0.3842
Epoch 11/50

413/413 ————— 1s 1ms/step - loss: 0.2970
- mae: 0.3791 - val_loss: 0.3250 - val_mae: 0.3951
Epoch 12/50

413/413 ————— 1s 1ms/step - loss: 0.2932
- mae: 0.3731 - val_loss: 0.3342 - val_mae: 0.4054
Epoch 13/50

413/413 ————— 1s 1ms/step - loss: 0.2892
- mae: 0.3700 - val_loss: 0.3221 - val_mae: 0.3884
Epoch 14/50

413/413 ————— 1s 1ms/step - loss: 0.2853
- mae: 0.3693 - val_loss: 0.3148 - val_mae: 0.3934
Epoch 15/50

413/413 ————— 1s 1ms/step - loss: 0.2845
- mae: 0.3692 - val_loss: 0.3124 - val_mae: 0.3885
Epoch 16/50

413/413 ————— 1s 1ms/step - loss: 0.2772
- mae: 0.3629 - val_loss: 0.3034 - val_mae: 0.3762
Epoch 17/50

413/413 ————— 1s 1ms/step - loss: 0.2781
- mae: 0.3638 - val_loss: 0.3306 - val_mae: 0.3835
Epoch 18/50

413/413 ————— 1s 1ms/step - loss: 0.2791
- mae: 0.3653 - val_loss: 0.3079 - val_mae: 0.3793
Epoch 19/50

413/413 ————— 1s 1ms/step - loss: 0.2727
- mae: 0.3601 - val_loss: 0.3076 - val_mae: 0.3840
Epoch 20/50

413/413 ————— 1s 1ms/step - loss: 0.2727
- mae: 0.3599 - val_loss: 0.3116 - val_mae: 0.3728
Epoch 21/50

413/413 ————— 1s 1ms/step - loss: 0.2702
- mae: 0.3575 - val_loss: 0.3069 - val_mae: 0.3742
Epoch 22/50

413/413 ————— 1s 1ms/step - loss: 0.2681
- mae: 0.3551 - val_loss: 0.3227 - val_mae: 0.3969
Epoch 23/50

413/413 ————— 1s 1ms/step - loss: 0.2668
- mae: 0.3554 - val_loss: 0.3058 - val_mae: 0.3721
Epoch 24/50

413/413 ————— 1s 1ms/step - loss: 0.2651
- mae: 0.3542 - val_loss: 0.2973 - val_mae: 0.3672
Epoch 25/50

413/413 ————— 1s 1ms/step - loss: 0.2659
- mae: 0.3545 - val_loss: 0.2968 - val_mae: 0.3816
Epoch 26/50

413/413 ————— 1s 1ms/step - loss: 0.2639
- mae: 0.3535 - val_loss: 0.3120 - val_mae: 0.3965
Epoch 27/50

413/413 ————— 1s 1ms/step - loss: 0.2596
- mae: 0.3503 - val_loss: 0.3040 - val_mae: 0.3747
Epoch 28/50

413/413 ————— 1s 2ms/step - loss: 0.2582
- mae: 0.3485 - val_loss: 0.3016 - val_mae: 0.3649
Epoch 29/50

413/413 ————— 1s 1ms/step - loss: 0.2578
- mae: 0.3488 - val_loss: 0.3015 - val_mae: 0.3742
Epoch 30/50

413/413 ————— 1s 1ms/step - loss: 0.2605
- mae: 0.3508 - val_loss: 0.3124 - val_mae: 0.3753
Epoch 31/50

413/413 ————— 1s 1ms/step - loss: 0.2764
- mae: 0.3514 - val_loss: 0.2973 - val_mae: 0.3715
Epoch 32/50

413/413 ————— 1s 1ms/step - loss: 0.2560
- mae: 0.3476 - val_loss: 0.2906 - val_mae: 0.3608
Epoch 33/50

413/413 ————— 1s 1ms/step - loss: 0.2515
- mae: 0.3437 - val_loss: 0.2840 - val_mae: 0.3566
Epoch 34/50

413/413 ————— 1s 1ms/step - loss: 0.2501
- mae: 0.3430 - val_loss: 0.2924 - val_mae: 0.3732
Epoch 35/50

413/413 ————— 1s 1ms/step - loss: 0.2494
- mae: 0.3419 - val_loss: 0.2854 - val_mae: 0.3707
Epoch 36/50

413/413 ————— 1s 1ms/step - loss: 0.2475
- mae: 0.3403 - val_loss: 0.2789 - val_mae: 0.3616
Epoch 37/50

413/413 ————— 1s 1ms/step - loss: 0.2474
- mae: 0.3402 - val_loss: 0.2843 - val_mae: 0.3631
Epoch 38/50

413/413 ————— 1s 1ms/step - loss: 0.2440
- mae: 0.3382 - val_loss: 0.3074 - val_mae: 0.4007
Epoch 39/50

413/413 ————— 1s 1ms/step - loss: 0.2472
- mae: 0.3405 - val_loss: 0.3060 - val_mae: 0.3758
Epoch 40/50

413/413 ————— 1s 1ms/step - loss: 0.2474
- mae: 0.3416 - val_loss: 0.2957 - val_mae: 0.3582
Epoch 41/50

413/413 ————— 1s 1ms/step - loss: 0.2430
- mae: 0.3382 - val_loss: 0.2822 - val_mae: 0.3625
Epoch 42/50

413/413 ————— 1s 1ms/step - loss: 0.2401
- mae: 0.3365 - val_loss: 0.2943 - val_mae: 0.3593
Epoch 43/50

413/413 ————— 1s 1ms/step - loss: 0.2411
- mae: 0.3377 - val_loss: 0.2803 - val_mae: 0.3574
Epoch 44/50

413/413 ————— 1s 1ms/step - loss: 0.2398
- mae: 0.3357 - val_loss: 0.2784 - val_mae: 0.3653
Epoch 45/50

413/413 ————— 1s 1ms/step - loss: 0.2369
- mae: 0.3340 - val_loss: 0.2788 - val_mae: 0.3586
Epoch 46/50

413/413 ————— 1s 1ms/step - loss: 0.2388
- mae: 0.3358 - val_loss: 0.2784 - val_mae: 0.3544
Epoch 47/50

413/413 ————— 1s 1ms/step - loss: 0.2359
- mae: 0.3334 - val_loss: 0.2804 - val_mae: 0.3628
Epoch 48/50

413/413 ————— 1s 1ms/step - loss: 0.2358
- mae: 0.3331 - val_loss: 0.2787 - val_mae: 0.3574
Epoch 49/50

413/413 ————— 1s 1ms/step - loss: 0.2349
- mae: 0.3320 - val_loss: 0.2823 - val_mae: 0.3552
Epoch 50/50

413/413 ————— 1s 1ms/step - loss: 0.2340
- mae: 0.3310 - val_loss: 0.2723 - val_mae: 0.3537

129/129 ————— 0s 915us/step - loss:
0.2672 - mae: 0.3496

Test Loss : 0.2671928107738495

Test MAE : 0.3495995104312897

129/129 ————— 0s 675us/step

RMSE : 0.5169069763906253

